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So… What Actually Wins? Feature Selection After EDA

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What Really Wins in March?

After completing the exploratory data analysis, one thing became very clear: not all Four Factors carry equal weight. Diff_eFG stood out immediately. With a correlation of approximately 0.66 and strong visual separation in the boxplots, shooting efficiency is the most powerful predictor of winning in this dataset. When teams consistently shoot better than their opponent, they win. It is not complicated. Efficient scoring drives outcomes.

Diff_FTR also showed meaningful predictive strength, with a correlation near 0.40. While not as dominant as effective field goal percentage, getting to the free throw line and converting those opportunities clearly improves win probability, especially in tight tournament games where small margins decide everything.

Diff_ORB and Diff_TOV were weaker individually, but neither appears to be noise. Offensive rebounding creates additional possessions, and turnover differential, though negatively correlated, reinforces a basic truth of basketball: giving the ball away lowers your chances of winning. Even if their standalone impact is smaller, they capture distinct elements of performance that may enhance predictive accuracy when combined with stronger variables.

For the logistic regression model, I will retain all four engineered features. Each represents a different dimension of team performance: shooting, ball security, rebounding, and free throw generation. However, based on both statistical magnitude and visual evidence, Diff_eFG is expected to carry the most influence in determining predicted win probability.

If there is one clear takeaway from this analysis, it is this: in March, shooting efficiency wins.